

Generation Interconnection Request System Impact Study Report

For Queue Project Q16-0599

Frederick Road 138 kV
January 22, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0599. The Interconnection Customer proposes a 105.7 MW solar photovoltaic facility interconnecting at the Frederick Road 138 kV substation in the Maryland area. The request was received on March 14, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 223 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$12,880,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Rio Blanco 2702 to Bruce Crossing 138 kV	Sierra Pacific Lumber Substation 138/1 kV XFMR	192.4	235	9.4
Alliance to Rio Blanco 2702 138 kV	Sierra Pacific Lumber 138/1kV Transformer	174.2	246	9.4
Bruce Crossing to Holopaw 138 kV	Sierra Pacific Lumber 138-1 kV Transformer	171.7	276	9.4
Mccall Substaion to Pebbledale 138 kV	Holopaw to Bruce Cossing 138 kV	127.2	352	5.1
Taber to Pebbledale 138 kV	Bruce Crossing to Holopaw 138 kV	109.3	450	5.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Rio Blanco 2702 to Bruce Crossing 138 kV	Rebuild line with higher-capacity conductor	\$2,645,000	100% (\$2,645,000)
Alliance to Rio Blanco 2702 138 kV	Rebuild line with higher-capacity conductor	\$1,135,000	100% (\$1,135,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Bruce Crossing to Holopaw 138 kV	Rebuild line with higher-capacity conductor	\$6,070,000	100% (\$6,070,000)
Mccall Substaion to Pebbledale 138 kV	Rebuild line with higher-capacity conductor	\$1,748,000	100% (\$1,748,000)
Pebbledale to Taber 138 kV	Reconductor line	\$1,282,000	100% (\$1,282,000)
	Total allocated to project		\$12,880,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.